Ali Lakzaei

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SUMMARY

- Privacy-preserving ML researcher designing ways for organisations to train models together without sharing raw data; work funded by MCI (**1 of 52** nationwide) and accepted at RESET 2026.
- Experienced in robust and trustworthy AI, from adversarial evaluation of vision transformers in a National Elite Foundation programme to machine learning detectors for attacks on smart grids.
- Data scientist building data pipelines and LLM agents for client companies, and teaching assistant for nine university courses.

EDUCATION

2023–2026 **M.Sc. in Computer Engineering**, Cybersecurity focus — GPA 18.93/20
Tarbiat Modares University — Ranked 1st in cohort, approx. cum laude
Thesis: A Blockchain-based Architecture for Vertical Federated Learning
Supervisor: Dr. S. Dorri Nogoorani · Advisor: Dr. H.R. Baghaee

2019–2023 **B.Sc. in Computer Engineering** — GPA 17.20/20
University of Tehran — Top 10% of cohort
Project: Investigating Patterns in Scraped Twitter Data Through Optimized NoSQL Querying
Supervisor: Dr. K. Fouladi

HONORS & AWARDS

2025 M.Sc. Thesis Grant – Mobile Communication Company of Iran, 1 of 52 nationwide

2025 Research Team Member, Ahmadi Roshan Program – Iran's National Elite Foundation

2024 Vezvaei Program Grant – Iran's National Elite Foundation

2023 Full Tuition Waiver – competitive admission to a state-funded M.Sc. program

RESEARCH INTERESTS

Privacy-Preserving Machine Learning
Vertical federated learning across organisations

LLM Agents
Agentic workflows over private, on-premise data

RESEARCH EXPERIENCE

2024–Present **Research Assistant, Blockchain Lab**
Tarbiat Modares University · *Supervisor: Dr. S. Dorri Nogoorani*
- Designed a blockchain-based architecture that rewards vertical federated learning members by their verified contribution, using zero-knowledge proofs and smart contracts on Hyperledger Besu (M.Sc. thesis, MCI-funded) [code]
- In collaboration with Dr. H.R. Baghaee, built a lightweight machine learning detector for false data injection attacks in smart microgrids, reaching 99.6% accuracy.
- Applied vertical federated learning to photovoltaic power forecasting in a simulated three-company setting, where it clearly reduced forecast error without sharing raw data (accepted at RESET 2026).

2025–2026 **Research Assistant, Ahmadi Roshan Program (National Elite Foundation)**
Amirkabir University of Technology · *Supervisor: Dr. M. Amirmazlaghani*
- Worked in a team of seven researchers developing a national system for evaluating the robustness and security of AI services, and designed a framework for testing models according to the level of access a service gives to the evaluator.
- Implemented white-box and black-box adversarial attacks on vision transformers in PyTorch and documented the findings in monthly technical reports.

2021–2023 **Research Assistant, Cyberspace Lab**
University of Tehran · *Supervisor: Dr. K. Fouladi*
- Built Python pipelines to collect, clean and analyse a large-scale Persian-language Twitter dataset using optimized NoSQL queries (B.Sc. final project).

PROFESSIONAL EXPERIENCE

2024–Present **Data Scientist**
Raydad Co. · *Data solutions company, Tehran*

- Built a pipeline that cleans multi-source client data and moves it from MongoDB to PostgreSQL, then analysed it with Python, Superset and Metabase to show clients their strengths and weaknesses.
- Developed and deployed (Docker) a chat-based LLM agent that queries company databases and runs web searches, evolving it from fixed function routing into a multi-agent LangGraph workflow.

PUBLICATIONS

[1] **A. Lakzaei**, S. Dorri Nogoorani, and H. R. Baghaee, "Vertical federated learning for photovoltaic power forecasting without sharing raw data," in *Proc. 2026 Conf. Renewable Energy Syst. Emerg. Technol. (RESET)*, Qom, Iran, Nov. 2026, accepted (to appear in IEEE Xplore).

[2] **A. Lakzaei** and K. Fouladi, "Relation between governance domain and cyberspace," in *Proc. 2nd Nat. Conf. Cyberspace (CYSP)*, Tehran, Iran, Jul. 2024 (in Persian).

[3] **A. Lakzaei**, A. Khanjari Mazrae, H. R. Baghaee, and M. K. Sheikh-El-Eslami, "Market clearing mechanism analysis for decentralized P2P energy trading by blockchain technology implementation," manuscript to be submitted. [demo] [code]

[4] K. Pourramezani, **A. Lakzaei**, H. R. Baghaee, and G. B. Gharehpetian, "AI-based cyber–physical detect–locate–recover framework for FDI cyberattack resilience and vulnerability assessment in μPMU-enabled microgrids: A data-driven governance-informed co-simulation," manuscript to be submitted.

TEACHING EXPERIENCE

2026 **Teaching Assistant, Cyber-Physical Systems** (graduate) Tarbiat Modares University

- Advised students in meeting sessions on the AI and blockchain parts of their course projects, as part of the TA team.

2025 **Teaching Assistant, Applied Machine Learning** (graduate) Tarbiat Modares University

- Designed projects and seminar topics tailored to each student's thesis proposal, and assessed students through their project presentations.

2025 **Teaching Assistant, E-Commerce Security** (graduate) Tarbiat Modares University

- Designed and graded course projects and led problem-solving sessions.

2024 **Teaching Assistant, Data Privacy & Security** (graduate) Tarbiat Modares University

- Graded students' homework and course projects.

2021–22 **Teaching Assistant, Advanced Programming** (B.Sc.) University of Tehran

- Designed programming projects, assessed them through student presentations, and taught live-coding classes.

TECHNICAL SKILLS

Machine Learning: PyTorch, scikit-learn, pandas, timm, Hugging Face
LLMs & Agents: LangGraph, LangChain, LangSmith, Ollama, MCP
Privacy & Blockchain: vertical federated learning, ZK proofs (Circom, Groth16), Solidity, Hyperledger Besu & Fabric, Hardhat
Data Engineering: PostgreSQL, MongoDB, ETL pipelines, Superset, Metabase
Programming & Tools: Python, SQL, C++, Git, Docker, Linux, LaTeX

LANGUAGES

Persian (native) English (C1 – IELTS Academic **8.0**: L 9.0, R 8.5, W 7.0, S 7.0)

REFERENCES

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